



Network-wide signal timing stochastic simulation optimization with environmental concerns

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HIGHLIGHTS

- Bi-objective stochastic simulation-based optimization (BOSSO) method is introduced.
- Bi-objective calibration and signal timing problems under uncertainties are well solved.
- BOSSO outperforms three other counterparts from various aspects.

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ABSTRACT

This study addresses a network-wide signal timing optimization problem with environmental concerns by a bi-objective stochastic simulation-based optimization (BOSSO) method. In this method, the global samples evaluated by costly simulation are used to build a type of surrogate model named the regressing Kriging model, which are then employed to predict bi-objectives of untested samples or filter noises from the evaluated samples. An adaptive selector is incorporated to determine which samples in the local trust-region are evaluated by costly simulation and which ones by the built regressing Kriging model. This helps to balance computational costs and accuracies of three quadratic regression models, especially when the variable dimension is high. The non-interactive role of a decision maker is taken to generate more non-dominated solutions around the desired bi-objective point. In the field experiments, an urban road network with 15 signalized and five non-signalized intersections in Changsha, China, is modeled as the simulation scenario by VISSIM. Then, the traffic simulation model is firstly calibrated from two aspects by the BOSSO method, which can well reproduce the reality. After that, the network-wide bi-objective signal timing optimization problem is also solved by the BOSSO method. Numerical results show that compared with the real-field traffic states, the total delay and vehicular emissions are reduced by at most 16.90% and 32.22% respectively under the budgeted number of simulations. Balance analyses also show the existence of a competing relationship between bi-objectives. Finally, the BOSSO method is validated to outperform three other counterparts (NSGA-II, BOTR and BOEGO) from various aspects.

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1. Introduction

Increasingly stringent regulations are being adopted worldwide to monitor and control urban traffic emissions. Notably, Frey et al. found that vehicular emissions are substantially influenced by traffic signalization primarily because of the accelerations that occur after a red light, rather than because of deceleration or idling [1]. Unal et al. unearthed that vehicular emission rates are highest during acceleration and tended to decrease (in descending order) for cruise, deceleration, and idle [2]. Rakha and Ding stated that the fuel consumption and emission rate increases considerably

if vehicle stops are introduced [3]. Skabardonis et al. [4,5], and Shabihkhani and Gonzales [6] provided a more accurate estimation of vehicular emissions with some analytical models that could estimate the distribution of time spent on each driving mode based on loop detector data. The signal control plan can greatly influence the fuel consumption and vehicular emissions on roads with signalized intersections, due to its significant impact on various driving modes and their durations. Thus, one alternative way of mitigating urban vehicular emissions is to optimize the signal control plan for intersections.

Previous vehicular emission models can be classified into two types, i.e., macroscopic and microscopic. Macroscopic emission models calculate emissions and fuel consumption of traffic flow based on the macroscopic traffic data. For example, the representatives of average-speed-based emission model [7,8] include

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MOBILE [9], EMFAC [10], COPERT [11], and VT-macro [12]. Integrated with meso- and macro-traffic models, these emission models have strong applicability to estimate regional or urban vehicular emissions and to forecast air quality. However, they do not consider the significant impact of time-varying speed and acceleration on vehicular emissions, resulting in a relatively low accuracy of estimation. While, microscopic emission models compute pollutant discharge and fuel consumption per second using instantaneous speed, acceleration and other parameters (e.g., vehicle type, road-grade). Their classic ones involve comprehensive modal emissions model (CMEM) [13], MOVES [14], IVE [15], and VT-Micro [16]. In contrast to their macroscopic counterparts, these models provide a higher-resolution description of emissions, but are time-consuming, data-intensive and mostly limited to the analysis of small urban areas.

Moreover, techniques that integrate vehicular emission models and traffic models result in simulation-based emission evaluation models. Such integrated models have attracted considerable attention due to their capabilities of evaluating the environmental impact of traffic management and control schemes. Some typically integrated models can refer to the literature reviews in [17, 18]. In addition, there are also other kinds of microscopic emission model-based integrated models including integrating CMEM with PARAMICS [19], integrating CO₂ and NO_x emission model VERSIT+ with PARAMICS [20], coupling the microscopic emission model HBEFA with the microscopic traffic simulator SUMO [21], the modal emission model combined with the cell transmission model (CTM) [22], and the integrated flow-emission model generated by combining a microscopic emission model with macroscopic flow model [23], etc.

With respect to simulation-based traffic signal control optimization with only consideration of vehicular emissions and (or) fuel consumption, Liao and Machemehl [24], and Park et al. [25] proposed signal timing optimization models for the minimization of vehicular emissions and fuel consumption based on microscopic traffic simulation. Hirschmann and Fellendorf [26] utilized a developed simulation toolbox to estimate pollutant emissions under various signal control schemes, and found that 5%–12% emission reductions depended on the types of pollutant and signal control scheme. Olivera et al. [21] used Particle Swarm Optimization (PSO) to optimize the traffic signal timing plan based on microscopic simulation, which reduced CO and NO_x by 23.3% and 29.3% respectively. Rakha et al. found that efficient coordinated signal control plan could reduce emissions up to 50% [27], and Madireddy et al. examined a signal coordination scheme that could reduce CO₂ and NO_x emissions by about 10% on an urban arterial road [20]. Both studies are based on traffic simulation as the evaluation tools.

Furthermore, regarding simulation-based traffic signal control optimization with both concerns of traffic efficiency and emissions, Stevanovic et al. [28] built a simulation-based optimization (SO) platform to investigate the network-wide traffic signal control optimization with considerations of various objectives. It was found that the total delay could be reduced by 11.94%, the fuel consumption by 3.84% and CO₂ emissions by 1.38%. Osorio and Nanduri [17] successfully optimized the regional intersection signal timing plan with environmental considerations in the SO framework. The total fuel consumption was calculated by an integrated high-resolution traffic and emission model, and then combined with the average travel time into a single objective. The similar procedure was also applied to another single objective function, which was a linear combination of expected travel time and total emission of four pollutants [18]. However, aforementioned studies lack the balance analyses of bi-objectives. To the best of our knowledge, the only work that has involved the trade-off analyses between traffic efficiency and emissions was done by Zhang et al. [22]. That is, a signal timing optimization model was formulated to simultaneously minimize the total delay and mean excess exposure, and its

Pareto optimal solutions were solved via a simulation-based bi-objective GA. Meanwhile, these Pareto optimal solutions showed the existence of competing relationship between the mean excess exposure and total delay.

In this study, our proposed bi-objective stochastic simulation-based optimization (BOSSO) approach [29] will be introduced to solve a network-wide signal timing optimization problem with the balance consideration of total delay and emissions under stochastic environments. The remainder of this paper is organized as follows. Section 2 reviews the SO methods. Section 3 formulates the network-wide bi-objective signal timing optimization problem and introduces the BOSSO method. Section 4 presents a real world application study and discussions on experimental results. Lastly, a brief conclusion is given in Section 5.

2. Literature review about SO methods

Due to the insufficiency of analytical models to describe the dynamic, stochastic, non-linear and complex urban road traffic systems, the mature traffic simulation software and secondary development technologies become an alternative way to flexibly design and simulate the sophisticated interactions between various traffic elements. This contributes to the accurate description of the complicated mapping relationship between signal control plan and network traffic state. However, the use of traffic simulation as an evaluation tool has the following drawbacks. Traditional optimization methods designed for noise-free optimization problems is out of work due to the deviation in objective values caused by many stochastic processes in the traffic simulation. It is unaffordable to use the computationally expensive simulation together with optimization algorithms that rely on extensive exploration of entire domain, e.g., Genetic Algorithm (GA), Tabu search, etc. The “Black-box” traffic simulator makes it difficult to derive explicit expression that fits classic optimization formulations (e.g., gradient descent).

Fortunately, many SO methods proposed in the past decades [30–36] are just applicable to address these weaknesses aforementioned. Here, only surrogate model-based SO techniques are specifically introduced. The critical step is to construct an analytical approximation of simulation-based components of an optimization problem (e.g., objective function, constraints), with the limited evaluated sample information. According to Søndergaard [37], the surrogate models can be classified into physical and functional types. The former type is application- (or problem-) specific, which has a physical (or structural) interpretation, and can provide the analytically tractable and global information. The latter type can provide a detailed local approximation of any kind of objective function or constraint, which depends on its analytical tractability. Another type of combined surrogate model proposed by Osorio and Bierlaire [38] takes full advantages of both physical and functional surrogate models. However, it is usually difficult to derive a physical surrogate model (e.g., an analytical and differentiable macroscopic traffic model) to approximate a real objective function or constraint under a complex scenario. Thus, it is popular to use the functional surrogate model instead of the physical one or the combined one to build the approximation. The common functional surrogate models include quadratic polynomials [39–43], spline models [34], radial basis functions (RBFs) [44–48], Kriging models [49,50], and support vector machines [51]. For more SO methods based on functional surrogate models can refer to references [42,43,51–53].

Moreover, there are usually three popular ways to deal with the multi-objective SO problems in traffic fields. (I) The weighting method is used to formulate multi-objectives into one weighted objective, which makes the multi-objective SO problem be converted to the single-objective SO problem. Then, a single-objective

SO method is employed to search the global optima. One can refer to some related studies [17,18,54]. However, these studies lack the balance analyses of various objectives. (II) Multi-objective evolutionary algorithm (EA) is also another alternative way. For example, in the SO framework, the multiple-objective GA was employed to optimize the signal plans so as to balance traffic efficiency and safety [55], and to solve the signal control optimization problem with the balance consideration of total delay and vehicular emissions [22]. Using Non-dominated Sorting Genetic Algorithm II (NSGA-II), Meng and Khoo [56] solved the dynamic ramp metering control optimization problem with both concerns of total delay and fairness index. To save computational costs, surrogate model based EAs [57] become promising methods to address (multi-objective) traffic SO problems. (III) With limited sample information evaluated by costly simulation, one surrogate model is constructed for each objective function, and intuitive infill strategies are designed to search new solutions to dominate at least one current non-dominated solution. Then, the Pareto front is updated and improved. For instance, to solve the bi-objective time-varying pricing problem, He et al. [53] used the probability of improvement (PI) and expected improvement (EI) to search a new improved solution based on regressing Kriging models built for bi-objectives. This algorithm is called Bi-objective Efficient Global Optimization (BOEGO) here. In addition, some other surrogate model based SO methods [43,58,59] can also be transplanted to solve similar multi-objective traffic SO problems.

However, there are still very limited methods that can efficiently address the multi-objective SO problems with stochastic noises and high dimensions. This is also the motivation of introducing the BOSSO method here to calibrate a large-scale traffic simulation model and solve the signal timing optimization problem with the balance consideration of vehicular emissions and total delay under uncertainties.

3. Methodology

3.1. Bi-objective signal timing stochastic simulation-based optimization model

In this section, traffic emissions will be calculated by incorporating the microscopic emission model proposed by Panis et al. [60] with the microscopic traffic simulator VISSIM. The reasons can be analyzed as follows. The microscopic emission model and high-resolution vehicular data from VISSIM contributes to the accurate estimation of vehicular emissions in the signalized road network. Meanwhile, VISSIM offers a user friendly graphical interface (GUI), and the geometry of any type of road networks can be designed easily and simulations can be set up simply. Using the COM interface, the user can manipulate the attributes of most internal objects dynamically, and realize the inter-process communication. This helps to bridge data exchange channel between VISSIM and Matlab, and then construct the SO platform. The following will introduce two evaluation indices about vehicular emissions and total delay, and then put forward the bi-objective signal timing stochastic SO model.

Vehicular Emissions: A linear combination of four pollutants' emissions (denoted as f^{CO_2} , f^{NO_x} , f^{VOC} and f^{PM} respectively) is given as

$$E = \frac{w^{CO_2}}{n^{CO_2}} f^{CO_2} + \frac{w^{NO_x}}{n^{NO_x}} f^{NO_x} + \frac{w^{VOC}}{n^{VOC}} f^{VOC} + \frac{w^{PM}}{n^{PM}} f^{PM} \quad (1)$$

where w^{CO_2} , w^{NO_x} , w^{VOC} and w^{PM} are the economic weighting parameters of CO_2 , NO_x , VOC and PM pollutants, respectively. n^{CO_2} , n^{NO_x} , n^{VOC} and n^{PM} are normalization constants, which are computed through the calibrated traffic simulator with the field implemented signal plan. The derivation process of the formulation

(1) and related parameter values can refer to Mayeres et al. [61], and Osorio and Nanduri [18]. The total emission (g) of pollutant $k \in \{CO_2, NO_x, VOC, PM\}$ during simulation is calculated by

$$f^k = \sum_{j=1}^{N_v} \sum_{t=t_j}^{T_j} ER_j^k(t) \quad (2)$$

where $ER_j^k(t)$ is the instantaneous emission rate of pollutant k (g/s) for vehicle j at time t , t_j and T_j are respectively the beginning and ending time-instants for vehicle j appearing in the simulation network, and N_v is the total number of vehicles during simulation. Moreover, $ER_j^k(t)$ is computed by the microscopic emission model in Panis et al. [60] based on a vehicle's instantaneous speed and acceleration, and formulated as

$$ER_j^k(t) = \max \{ E_{0j}^k; C_{1j}^k + C_{2j}^k v_j(t) + C_{3j}^k v_j(t)^2 + C_{4j}^k a_j(t) + C_{5j}^k a_j(t)^2 + C_{6j}^k v_j(t) a_j(t) \} \quad (3)$$

where $v_j(t)$ is the instantaneous speed (m/s), $a_j(t)$ is the instantaneous acceleration (m/s^2), E_{0j}^k is the minimal emission rate of pollutant k (g/s) for vehicle j , and C_{1j}^k to C_{6j}^k are emission rate coefficients for pollutant k and vehicle j . For certain pollutants (e.g., NO_x and VOC), variant functional forms (reflected by different emission rate coefficients) are derived for acceleration mode (defined as $a_j(t) \geq -0.5 m/s^2$) and deceleration mode (with $a_j(t) < -0.5 m/s^2$). According to Panis et al. [60], the minimal emission rate E_{0j}^k is equal to zero for all pollutants and vehicles, and these emission rate coefficients are the same for all vehicles.

Total Delay: the relative total delay is utilized to evaluate the traffic efficiency, which is formulated as

$$D = \left(\sum_{j=1}^{N_v} y_j \right) / C \quad (4)$$

where y_j is the delay experienced by vehicle j during simulation. C is the normalization constant about the total delay, which is calculated by the calibrated traffic simulator under the field implemented signal plan.

Under stochastic simulation environments, one signal timing plan x will result in various E and D values. Then, the bi-objective signal timing stochastic SO problem is expressed as

$$\text{Minimize}_{x \in \Omega} f(x, \varsigma) = \{E[E(x, \varsigma)], E[D(x, \varsigma)]\} \quad (5)$$

where $\Omega = \{x \in \mathbb{Z}^n: \mathbf{a}_1 \leq x \leq \mathbf{b}_1\}$ is the decision space, $\mathbf{a}_1$ and $\mathbf{b}_1$ respectively denote the lower and upper bounds of decision vector x with the dimension n . ς denotes the stochastic exogenous parameters in the traffic simulator. $E(x, \varsigma) = e(x, \varsigma) + \delta_e$, $\delta_e \sim N(0, \sigma_e^2)$ and $D(x, \varsigma) = d(x, \varsigma) + \delta_d$, $\delta_d \sim N(0, \sigma_d^2)$ denote stochastic bi-objective values output from the simulation. $e(x, \varsigma)$ and $d(x, \varsigma)$ are true expected bi-objective values. Two random variables δ_e and δ_d are assumed to follow the normal distribution. $E[\cdot]$ represents the expectation operator. Apparently, the formulation (5) aims to simultaneously minimize the expectations of vehicular emissions and total delay. Due to the discrete integer type of x , the difficulty of solving the problem (5) increases as n gets larger.

Moreover, for a given x and ς , a larger number of simulation repetitions are desired to smooth the noises better and make the average bi-objective values closer to their expectations, which however would greatly increase additional computational costs. Thus, the average approximation method (AAM) [62] is not an ideal choice to address the random noises in the formulation (5). Inspired by the studies in [51,53], the following BOSSO method will take full advantage of the regressing Kriging model to deal with the stochastic noises.

3.2. Bi-objective stochastic simulation-based optimization method

The section introduces our proposed BOSSO method [29], which improves a bi-objective derivative-free trust region (BOTR) algorithm [43] from three aspects. (1) The regressing Kriging model [53, 63] is built with global simulated samples, and then used to predict (re-estimate) bi-objective values of untested (simulated) samples in the local trust region. This mitigates the negative effect of stochastic noises in searching the expected bi-objective values, and meanwhile save a lot of computation costs. (2) The adaptive selector is set to determine which samples in the local trust region are evaluated by costly simulation, and which ones are predicted by the built regressing Kriging model. This helps to balance the computational costs and accuracies of three quadratic regression models. (3) The non-interactive role of a decision maker is taken to influence the search direction and trust-region, which benefit to guiding one part of non-dominated solutions towards his/her desired bi-objective values. In this condition, the BOSSO method is capable to address the bi-objective stochastic SO problems with high dimensions and the decision maker's preference. The main steps of BOSSO are presented in the following.

Initialization: Set the initial point $x^{(0)}$ with the dimension n , and evaluate the bi-objective function values $F(x^{(0)}) = [f_1(x^{(0)}), f_2(x^{(0)})]$. Set the initial set of non-dominated solutions $\chi^{(0)} = \{x^{(0)}\}$. The initial trust-region radius is $\Delta^{(0)}(x^{(0)}) \in (0, \Delta_{\max}]$. The maximum and tolerable trust-region radiuses are $\Delta_{\max} > 0$ and $\Delta_{\text{tol}} \geq 0$, respectively. The minimum closeness allowed between bi-objective values is $\delta > 0$. The critical reduction ratio is $\eta \in [0, 1)$. The decrease and the increase parameters about the trust-region are set as $\tau_{\text{dec}} \in (0, 1)$ and $\tau_{\text{inc}} > 1$, respectively. Two constant coefficients about the norm of the gradient of quadratic regression model are $\mu > \beta > 0$, and the reduction coefficient about the trust-region in the inner iteration is $\omega \in (0, 1)$. The sets of all simulated and filtered samples are $Z = \emptyset$ and $X = \emptyset$, respectively.

Algorithm: For each **outer iteration** $k = 1, 2, \dots$,

Step 1 (Iteration Determination): Return the set of non-dominated points $\chi^{(k-1)}$ from the previous iteration. Select the most isolated point $x_c^{(k)}$ with the trust-region radius $\Delta^{(k-1)}(x_c^{(k)})$ from $\chi^{(k-1)}$ (c.f., Appendix). Then, set the reference point $r^{(k)} = F(x_c^{(k)})$, $Y^{(k)} = \emptyset$, and $\Delta_0^{(k)} = \Delta^{(k-1)}(x_c^{(k)})$.

Step 2 For each **inner iteration** $j = 1, 2, \dots$,

Step (i) (*Determine the current samples to be evaluated by costly simulation*): Set $\tilde{\Delta}^{(k)} = \omega^{j-1} \cdot \Delta_0^{(k)}$, employ Latin Hypercube Sampling (LHS) to generate a poised set of samples $S_p = \{y_1^{(j)}, y_2^{(j)}, \dots, y_d^{(j)}\}$ in the trust-region $B(x_c^{(k)}, \tilde{\Delta}^{(k)})$. Use an adaptive selector to select d_c samples $S_c = \{y_1^{(j)}, y_2^{(j)}, \dots, y_{d_c}^{(j)}\}$ randomly from S_p , which will be evaluated by the costly simulation, and $d_c = \min \left\{ d_{\max}, \max \left(d_{\min}, \left\lfloor \log(k) / (c_f \cdot (\tilde{\Delta}^{(k)})^2) \right\rfloor \right) \right\}$, where $c_f > 0$. Evaluate the selected d_c samples by the costly simulation, and add these evaluated samples to Z : $Z \leftarrow Z \cup S_c$.

Step (ii) (*Prediction and re-estimation by global regressing Kriging models*): Employ Z to construct two regressing Kriging models for bi-objectives, which are then used to predict the bi-objective values of unevaluated samples in S_p (i.e., $S_p \setminus S_c$), and re-estimate the samples in S_c to filter stochastic noises from simulations. Add the filtered samples to $Y^{(k)}$ and X : $Y^{(k)} \leftarrow Y^{(k)} \cup S_c$, and $X \leftarrow X \cup S_c$.

Step (iii) (*Construct the quadratic regression models*): Based on the poised sample set S_p , and their predicted and filtered function

values of f_1 and f_2 , construct the quadratic regression model $\tilde{m}_i^{(j)}$ of $f_i^{(k)}: \tilde{m}_i^{(j)}(x_c^{(k)} + s) = \tilde{c}_i^{(j)} + s^T \tilde{g}_i^{(j)} + \frac{1}{2} s^T \tilde{H}_i^{(j)} s$, $i = 1, 2, 3$, where $\tilde{g}_i^{(j)}$ and $\tilde{H}_i^{(j)}$ are the gradient and Hessian of $\tilde{m}_i^{(j)}$ at $s = 0$. Here, the formulations of $f_3^{(k)}$ and $m_i(x)$ can refer to the reference [43].

Step (iv) (*Criticality checking step*): If $\tilde{\Delta}^{(k)} \leq \max \left\{ \Delta_{\text{tol}}, \min_{i=1,2,3} \mu \left\| \tilde{g}_i^{(j)} \right\|, i = 1, 2, 3 \right\}$, then set $m_i^{(k)} = \tilde{m}_i^{(j)}$ and $\Delta_c^{(k)} = \min \left[\max \left\{ \tilde{\Delta}^{(k)}, \beta \min_{i=1,2,3} \left\| \tilde{g}_i^{(j)} \right\| \right\}, \Delta_0^{(k)} \right]$, and go to Step 3. Otherwise, increment j by one and go to Step (i).

Step 3 (Solve three single-objective problems): Solve each single-objective problem in the trust-region $B(x_c^{(k)}, \Delta_c^{(k)})$ and let $z_i^{(k)} = \text{argmin} \{m_i^{(k)}(x): x \in B(x_c^{(k)}, \Delta_c^{(k)})\}$, $i = 1, 2, 3$. Evaluate $z_i^{(k)}$ by the costly simulation, and add these three evaluated samples to Z : $Z \leftarrow Z \cup \{z_1^{(k)}, z_2^{(k)}, z_3^{(k)}\}$.

Step 4 (Update the regressing Kriging models): Re-build these two regressing Kriging models with Z , and use the updated regressing Kriging models to re-estimate these three samples $\{z_1^{(k)}, z_2^{(k)}, z_3^{(k)}\}$ so as to smooth the stochastic noises from the simulation evaluations. Add these three filtered solutions to X : $X \leftarrow X \cup \{z_1^{(k)}, z_2^{(k)}, z_3^{(k)}\}$.

Step 5 (Calculate the reduction ratios): Compute the reduction ratio for each newly evaluated point in $Y^{(k)}$, $\{z_1^{(k)}, z_2^{(k)}, z_3^{(k)}\}$ and $\chi^{(k-1)}$, and remove the samples with insufficient reduction ratio from $Y^{(k)}$.

Step 6 (Update the trust-region radiuses): Determine a trust-region radius for each newly evaluated point in $Y^{(k)}$, $\{z_1^{(k)}, z_2^{(k)}, z_3^{(k)}\}$ and $\chi^{(k-1)}$.

Step 7 (Update the set of non-dominated solutions): Compare the vector values $F(x) = [f_1(x), f_2(x)]$ for $x \in \chi^{(k-1)} \cup Y^{(k)} \cup \{z_1^{(k)}, z_2^{(k)}, z_3^{(k)}\}$, and give the updated set of non-dominated points $\chi^{(k)}$.

Step 8 (Iteration evaluation): Examine the successful iteration and check the termination condition.

Step 9 (Output the Pareto front and optimal solutions for Z and X)

Here, Step 5 to Step 8 accord with those in the BOTR algorithm [43].

4. Real-field experiments

This section will firstly employ the BOSSO method to calibrate the microscopic traffic simulation model from two aspects for a real-field study site, and then solve the network-wide signal timing stochastic SO problem with the balance consideration of vehicular emissions and total delay. Moreover, three other counterparts (i.e., NSGA-II, BOTR and BOEGO) will be used to optimize the network-wide signal timing plan, and then be compared with the BOSSO method in terms of convergence performance, diversity of solutions and computational costs. Finally, the reliability of the optimized signal plans by four algorithms will be validated under stochastic environments.

Before numerical experiments, two indices to judge the wellness of non-dominated solutions are introduced as follows. A revised Generational Distance (GD) is formulated to evaluate the closeness of the generated solutions to the meta-Pareto front, and a Hypervolume (HV) indicator is used to determine the diversity of these solutions. The former is based on the well-known measure

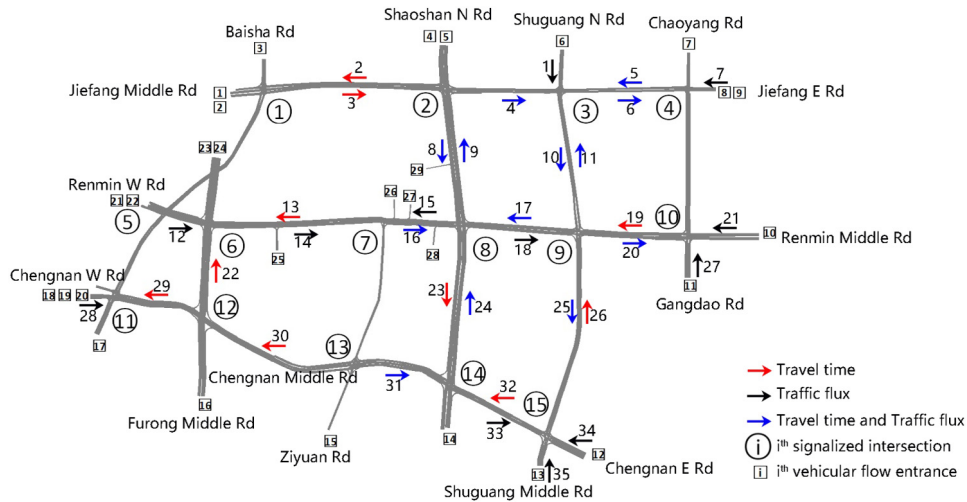


Fig. 1. VISSIM model of a regional network in Changsha, China.

GD [64], and is revised as

$$GD^* = \sqrt{\sum_{j=1}^v \|F(x_j) - H(x_j^*)\|^2} / v \quad (6)$$

where v denotes the number of the set of solutions (i.e., $S = \{x_1, x_2, \dots, x_v\}$) generated by one algorithm. If $x_j \in \Omega'$, then $x_j^* = x_j$ and $H(x_j^*) = F(x_j)$; else, $H(x_j^*)$ is the meta-Pareto efficient point that is the closest to $F(x_j)$. H denotes a meta-Pareto front and Ω' indicates its corresponding meta-Pareto solution set. For example, the meta-Pareto front is set as ultimate non-dominated solutions by one algorithm (c.f., Fig. 2) or those by various algorithms (c.f., Fig. 4(a)). The smaller GD^* indicates better convergence performance. If all solutions in the set S are meta-Pareto optimal, GD^* will become zero.

The value of HV is equal to the area enclosed by the generated solutions and a reference point (i.e., $R_H = [f_1(x_0), f_2(x_0)]$) that must be dominated by all $F(x_j)$, $j = 1, 2, \dots, v$ [43]. Thus, the larger HV is the better the diversity will be. With the solution set S sorted in the increasing order concerning f_1 , HV is computed as

$$HV = \sum_{i=0, \dots, v-1} \{f_1(x_0) - f_1(x_{i+1})\} \{f_2(x_i) - f_2(x_{i+1})\} \quad (7)$$

4.1. Construction of simulation model and its calibration

The microscopic traffic simulator VISSIM is firstly used to model one urban road network in Changsha, China as the experimental scenario, which includes 15 signalized intersections (with 47 signal phases), five non-signalized intersections, 29 entries, 24 exits, and 31 road links, c.f., Fig. 1. Then, in the SO platform that is built with VISSIM and Matlab through COM interface, the network-wide traffic simulation model has to be calibrated before the signal timing optimization. Here, the calibration parameters cover traffic flow inputs of 29 entrances and three driving behavior-related parameters in the Wiedemann74, i.e., the dimension of parameter vector $h = 32$. Moreover, the upper and lower bounds of 29 traffic flow inputs are set according to the designed levels of road links, and those of three driving behavior parameters accord with the default upper and lower values in VISSIM. The benchmark data for the calibration include the complete traffic volumes in 25 links (i.e., $N_f = 25$), which are collected near signalized intersections in the Scats systems, and the complete travel time data in 23 links (i.e., $N_t = 23$), which are extracted from taxis' GPS (c.f., Fig. 1). These two types of real-field data were collected during 9:00–9:30

AM on 2013/11/16 with time intervals of 5 and 2 min, respectively, which are then aggregated to denote the network-wide traffic states for this half an hour.

Different from the single-objective calibration model by Zhang et al. [65], which cannot comprehensively characterize the network-wide traffic states, the calibration here is to simultaneously minimize the differences between the field and simulated traffic states with respect to traffic volume and travel time. Hence, this bi-objective calibration model under stochastic environments is formulated as

$$\text{Minimize } f(p, \varsigma) = \{E[A_f(p, \varsigma)], E[A_t(p, \varsigma)]\} \quad (8)$$

where p indicates the vector of calibration parameters, the decision space is $\Omega_2 = \{p \in \mathbb{R}^h: a_2 \leq p \leq b_2\}$, a_2 and b_2 denote the lower and upper bounds of calibration parameters, respectively. $A_f(p, \varsigma) = (\sum_{i=1}^{N_f} |F_i - FS_i(p, \varsigma)| / F_i) / N_f$ and $A_t(p, \varsigma) = (\sum_{j=1}^{N_t} |T_j - TS_j(p, \varsigma)| / T_j) / N_t$ indicate the Mean Absolute Percentage Error (MAPE) about traffic volume and travel time, respectively. F_i and $FS_i(p, \varsigma)$ indicate the field and simulated traffic volume in link i , respectively. T_j and $TS_j(p, \varsigma)$ denote the field and simulated travel time in link j , respectively. During the calibration experiment, $f_d = (0.2, 0.2)$ in A_f - A_t plane is set as the desired point in Appendix. Other critical parameters of the BOSSO method are as follows: $d_{max} = 80$, $d_{min} = 40$, $\Delta^{(0)} = 60$, $\Delta_{max} = 100$, $\Delta_{tol} = 20$, $\delta = 0.01$, $\eta = 10^{-8}$, $\tau_{dec} = 0.6$, $\tau_{inc} = 1.05$, $\mu = 8.0 \times 10^{14}$, $\beta = 6.0 \times 10^{14}$, $\omega = 0.6$, $c_f = 6.0 \times 10^{-5}$, and the budgeted simulation time is 1000. Because each simulation for this half an hour of real traffic costs about 90 s, it takes approximately 24 h to complete the BOSSO algorithm.

The ultimate non-dominated solutions in Fig. 2 are arranged in the ascending order concerning A_f to facilitate their indications in the following. Obviously, the 5th to 8th solutions are very close to the desired point f_d , i.e., $(A_f = 0.2, A_t = 0.2)$. Moreover, the 8th non-filtered solution in Fig. 2(a) (i.e., $(A_f = 0.2406, A_t = 0.1876)$) and the 8th filtered solution in Fig. 2(b) (i.e., $(A_f = 0.2450, A_t = 0.2099)$) are both the nearest to f_d , which correspond to the same calibration parameters and are considered satisfactory to reproduce the reality. Then, with the well calibrated traffic stochastic simulation model, the bi-objective evaluation values are collected from 300 repetitive simulations, and two mean values are calculated as $\bar{A}_f = 0.23097$ and $\bar{A}_t = 0.26206$, respectively. Apparently, the mean value point, i.e., $(\bar{A}_f = 0.23097, \bar{A}_t = 0.26206)$, is closer to the 8th filtered solution than to the 8th non-filtered solution.

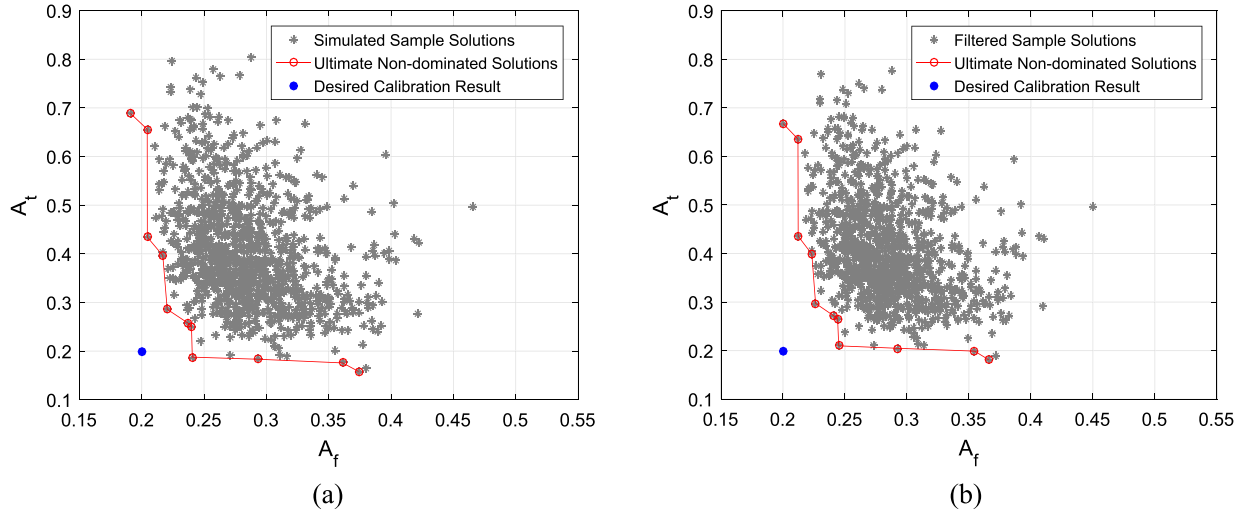


Fig. 2. Numerical solutions for calibration, (a) simulated solutions, (b) filtered solutions.

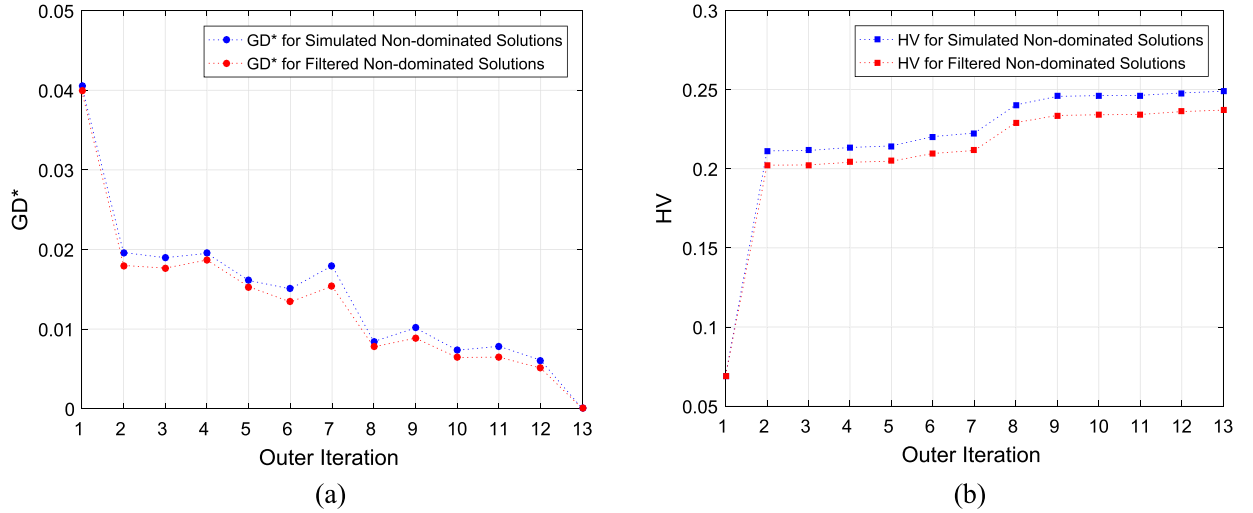


Fig. 3. Convergence process with respect to outer iteration, (a) GD^* , (b) HV. Note: $R_H = [0.55, 0.9]$ for calculation of HV.

This validates that the regressing Kriging model can efficiently filter noises from the simulated solutions. Meanwhile, Fig. 3(a) illustrates that the GD^* values for filtered solutions are usually smaller than those for simulated (i.e., non-filtered) solutions. It implies the well capability of regressing Kriging model in helping to search the closer non-dominated solutions to the meta-Pareto front, i.e., the better exploration capability. In addition, Fig. 3(b) demonstrates that the diversity of non-dominated solutions gets larger with the increase of outer iteration, and almost stabilizes after the 9th outer iteration, which shows the convergence of the BOSSO method.

4.2. Network-wide signal timing optimization

This section attempts to optimize the network-wide signal timing plan (i.e., green timings of 47 signal phases for 15 signalized intersections) with the calibrated traffic simulator as the evaluation tool. In the following experiments, the critical parameters of the BOSSO method are set as: $d_{max} = 100$, $d_{min} = 50$, $\Delta^{(0)} = 70$, $\Delta_{max} = 200$, $\Delta_{tol} = 20$, $\delta = 0.01$, $\eta = 10^{-8}$, $\tau_{dec} = 0.7$, $\tau_{inc} = 1.1$, $\mu = 8.0 \times 10^{16}$, $\beta = 6.0 \times 10^{16}$, $\omega = 0.7$, $c_f = 6.0 \times 10^{-5}$, and the budgeted number of simulations is 1000. The parameters involved in the calculation of E (i.e., Eq.(1)) can refer to Panis et al. [60].

The non-interactive role of a decision maker is removed so as to diversify the non-dominated solutions. Meanwhile, three other algorithms (i.e., NSGA-II, BOTR and BOEGO) are also introduced to be compared with the BOSSO method. Specifically, NSGAII and BOTR both employ AAM to filter the noises from simulated bi-objective values. This section only focuses on filtered non-dominated solutions due to their smaller closeness to the expected values, which has been validated in Section 4.1.

Fig. 4(a) illustrates that the ultimate non-dominated solutions are all in the left-bottom corner of the real-field values, which means the total delay and vehicular emissions under the field implemented signal plan are improved at various degrees by these four algorithms. The meta-Pareto front only consists of the non-dominated solutions by BOSSO, which shows the outperformance of BOSSO in searching more frontier solutions. Moreover, Fig. 4(b,c) demonstrates that GD^* and HV by BOSSO rapidly drops and increases respectively as the number of function evaluations grows, and they both have little change once the number of function evaluations reaches about 400. It implies BOSSO obtains the convergence and searches the well enough non-dominated solutions at the cost of only hundreds of simulations. While, NSGA-II, BOTR and BOEGO perform obviously worse than BOSSO, that is, their corresponding GD^* and HV values are respectively larger and smaller than those by BOSSO, and the convergence seems not

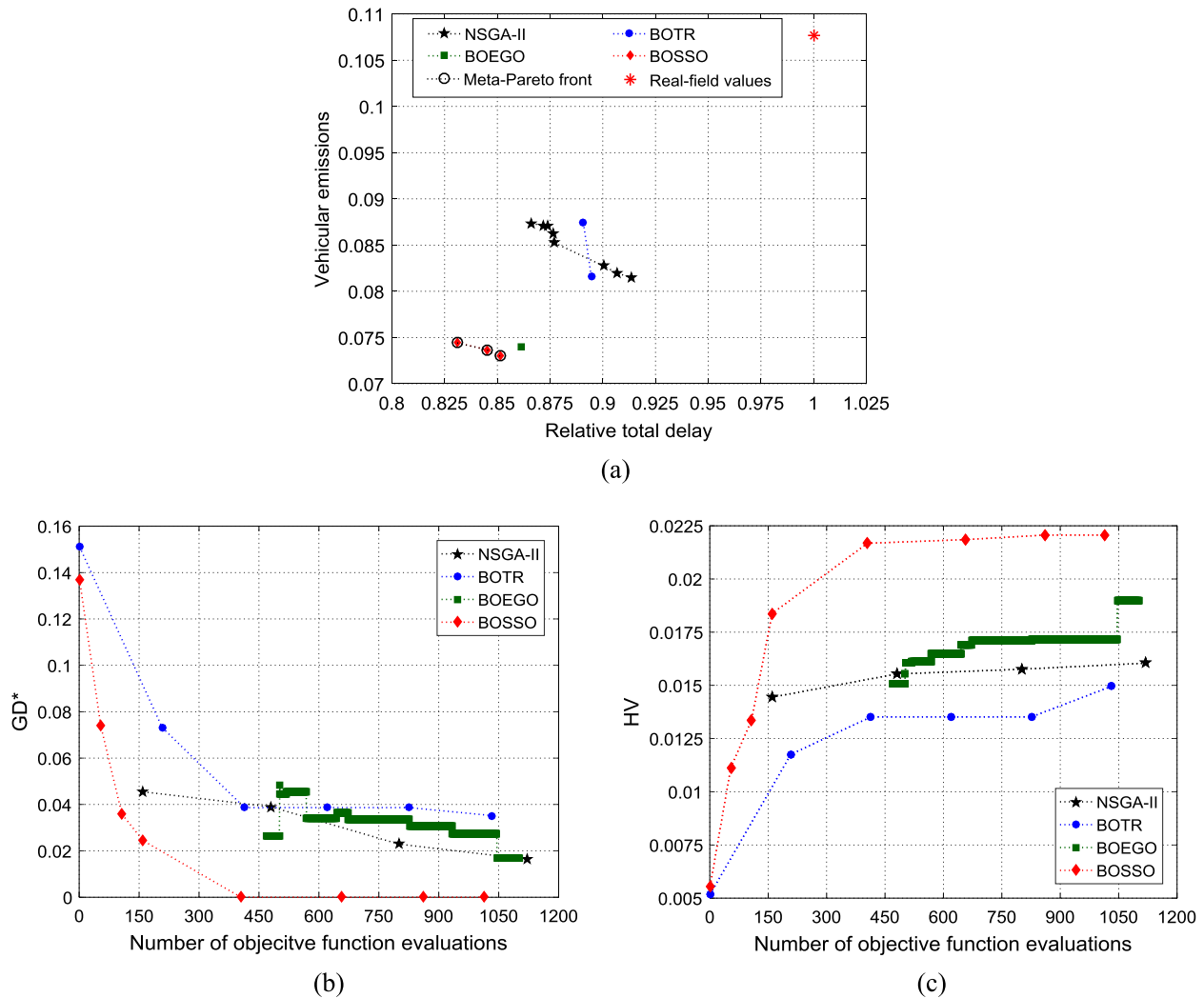


Fig. 4. Performance comparison of four algorithms, (a) Non-dominated solutions, (b) GD^* , (c) HV. Note: $R_H = [1.2, 0.13]$ for the calculation of HV.

Table 1
Detailed performance comparison of four algorithms.

	No. of FEs	No. of NSs	GD^*	HV	Computation time
NSGA-II	1120	8	0.0164	0.0160	28.0 h
BOTR	1032	2	0.0352	0.0150	25.8 h
BOEGO	1101	1	0.0168	0.0190	81.6 h
BOSSO	1014	3	0	0.0221	25.4 h

Note: No. of FEs and No. of NSs denote the number of function evaluations and non-dominated solutions, respectively.

reached. More detailed comparison can refer to Table 1. Therefore, it is drawn that with the limited computational costs, BOSSO can successfully search the more frontier and more diverse non-dominated solutions, compared with NSGA-II, BOTR and BOEGO.

After that, the balance analyses are carried out as follows. Table 2 lists three non-dominated solutions by BOSSO in the increasing order concerning the total delay. Taking the 1st solution as the benchmark, two absolute percentage changes respectively about total delay and vehicular emissions (i.e., P_d and P_e) are obtained for the other two solutions. In other words, when replacing the 1st solution with others, the reduction of vehicular emissions is paid off by the increase of total delay in a certain level. Moreover, P_d^+

and P_e^+ denote another two absolute percentage changes respectively about total delay and emissions, compared with two normalization constants (i.e., (1.0, 0.1077)). These two normalization constants are calculated by Eq. (4) and Eq. (1) respectively under the calibrated traffic simulator with the field implemented signal plan. Obviously, the total delay and vehicular emissions are both reduced significantly compared with the field bi-objective values.

4.3. Reliability analyses of various signal plans

This section will check the reliability of various non-dominated solutions by four algorithms. Besides of the field implemented signal plan, four other optimized ones are selected, i.e., the 1st one by BOSSO, the only one by BOEGO, the 2nd one by BOTR and the 6th one by NSGA-II. 300 repetitive simulations are run for each selected signal plan under stochastic environments. Fig. 5 illustrates their empirical cumulative distribution functions (CDFs) about the relative total delay and vehicular emissions. It is shown that the bi-objective values for four optimized signal plans are apparently improved compared with those for the field implemented one. The 1st solution by BOSSO and the only one by BOEGO perform better than the 2nd one by BOTR in both terms of relative total delay and vehicular emissions, which is followed by the 6th one by NSGA-II. Meanwhile, the non-dominated relationship between the 1st solution by BOSSO and the only one by BOEGO has also

Table 2
Three non-dominated solutions and the comparisons.

Signal timing plan	Relative total delay D	Emissions E	(P_d, P_e) (%)	(P_d^+, P_e^+) (%)
1	0.8310	0.0744	(0, 0)	(16.90↓, 30.92↓)
2	0.8449	0.0737	(1.67↑, 0.94↓)	(15.51↓, 31.57↓)
3	0.8512	0.0730	(2.43↑, 1.88↓)	(14.88↓, 32.22↓)

Note: ↓ and ↑ respectively indicates the reduction and increase compared with the benchmark solution.

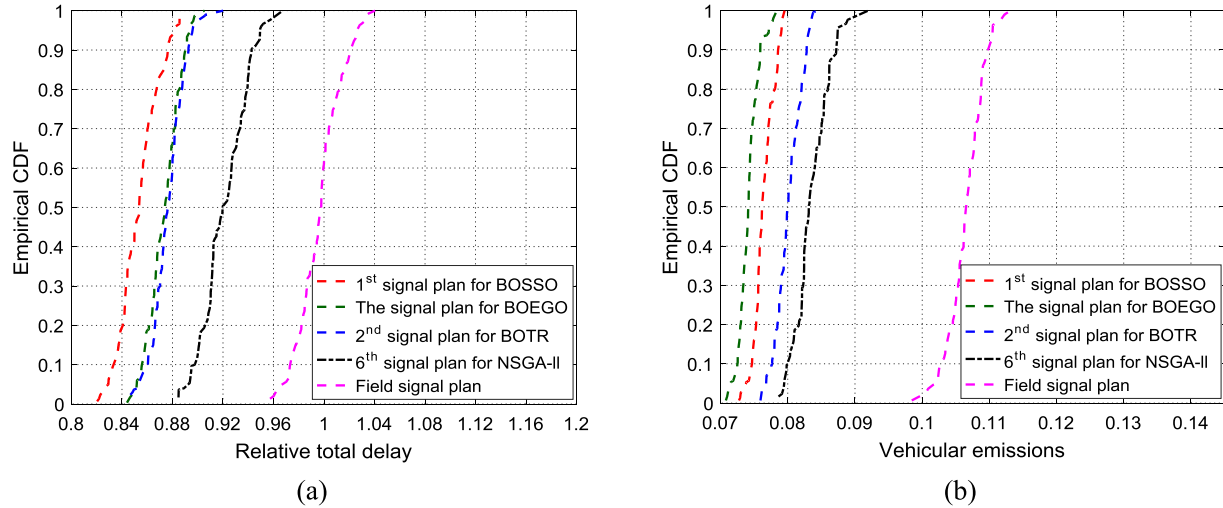


Fig. 5. Comparison of empirical CDFs with respect to (a) relative total delay, (b) vehicular emissions.

been validated. Thus, the non-dominated and dominated relationships among these five solutions are successfully validated under stochastic environments.

In conclusion, the total delay and vehicular emissions can be simultaneously improved when the signal timing plan is not good enough, e.g., the field implemented plan compared with the optimized ones, or the optimized plans by NSGA-II and BOTR compared with those by BOSSO. Meanwhile, the competing relationship between bi-objectives also exists. That is, the improvement of environmental benefits (traffic efficiency) sacrifices the traffic efficiency (environmental benefits), c.f., three non-dominated solutions in Table 2. In this condition, the preferred signal timing plan completely depends on the balance consideration of bi-objectives by the decision maker.

5. Conclusions

As more and more attentions paid to environmental concerns, traffic efficiency is treated as an evaluation indicator instead of the only indicator in the traffic management and control. This study proposes a network-wide signal timing stochastic SO model with the balance consideration of total delay and vehicular emissions. Meanwhile, the BOSSO method is introduced to address this type of high-dimensional, computationally expensive, and bi-objective stochastic SO problem. In the real world application, an urban road network with 15 signalized intersections and five non-signalized intersections in Changsha, China, is selected as the experimental scenario. Firstly, the bi-objective calibration problem (i.e., simultaneous minimization of two expectations of MAPEs about travel time and traffic flux) is solved by BOSSO. Results show that the calibrated network-wide traffic simulator can well reproduce the reality, and the regressing Kriging model efficiently mitigates noises from the simulated samples. Based on the calibrated traffic simulator, the network-wide signal timing plan is optimized by BOSSO to trade off the total delay and vehicular emissions. It is found that under the limited computational costs, the field total delay and

vehicular emissions are reduced by at most 16.90% and 32.22%, respectively. The evolution processes of GD^* and HV illustrate that BOSSO outperforms three other counterparts (NSGA-II, BOTR and BOEGO) in the convergence performance and searching more frontier and more diverse non-dominated solutions. Moreover, the balance analyses for three non-dominated solutions by BOSSO illustrate that the improvement of environmental benefits (traffic efficiency) sacrifices the traffic efficiency (environmental benefits). That is, the competing relationship between bi-objectives exists. Finally, through the repetitive simulations under stochastic environments, the empirical CDFs about the relative total delay and vehicular emissions are employed to successfully validate the reliability of optimized signal plans by these four algorithms.

In summary, the BOSSO method can well address the network-wide bi-objective calibration and signal timing optimization problems under uncertainties, which can also be easily transplanted to solve other similar problems. Future researches will focus on investigating the impact of the network-wide signal control coordination scheme on vehicular emissions and total delay. The multi-criteria (i.e., larger than 2)-based network-wide signal control SO problems and corresponding efficient SO algorithms will be elaborated in the future.

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Appendix. Procedure of selecting the most isolated point

The procedure of selecting the most isolated point $x_c^{(k)}$ from $x^{(k-1)}$ is presented as **Sub-Algorithm A1**, which helps to obtain more Pareto optimal solutions around the desired point f_d . The main idea is described as follows. Due to the non-interactive role of a decision maker, f_d guides the most isolated point towards its

neighborhood. Meanwhile, a poised set of samples will be generated in a trust-region centralized at the most isolated point. These samples will become Pareto optimal solutions if some conditions are satisfied. In this condition, more Pareto optimal solutions may exist around f_d , from which the decision maker can select the most satisfactory solution. Therefore, the first step is to determine whether there are solutions that dominate the desired point f_d in $\chi^{(k-1)}$, with the larger trust-region radius than the tolerable value. If not exist, the most isolated point is selected as the solution in $\chi^{(k-1)}$, which is the closest to f_d and has a larger trust-region radius than the tolerable value. Otherwise, the most isolated point is selected by the selection procedure in the reference [43].

Sub-algorithm A1: Select the most isolated point $x_c^{(k)}$

INPUT

- 1: The desired point of a decision maker f_d
- 2: The previous set of non-dominated solutions $\chi^{(k-1)}$
- 3: Set $\Theta = \emptyset$.

BEGIN

- 3: **For** each $x \in \chi^{(k-1)}$ and $\Delta^{(k-1)}(x) > \Delta_{tol}$ **do**
- 4: **If** $F(x) \succ f_d$ **then**
- 5: $\Theta \leftarrow \Theta \cup x$
- 6: **Else**
- 7: Θ is unchanged
- 8: **End if**
- 9: **End for**
- 10: **If** $\Theta = \emptyset$ **then**
- 11: $\gamma^{(k)}(x) = \|f_d - F(x)\|$, where $x \in \chi^{(k-1)}$ and $\Delta^{(k-1)}(x) > \Delta_{tol}$;
 $\gamma_{\min}^{(k)} = \min \{ \gamma^{(k)}(x) \mid \Delta^{(k-1)}(x) > \Delta_{tol}, x \in \chi^{(k-1)} \}$;
 $\hat{x} = \arg \min \{ \gamma^{(k)}(x) \mid \Delta^{(k-1)}(x) > \Delta_{tol}, x \in \chi^{(k-1)} \}$;
 $x_c^{(k)} = \hat{x}$ (c.f., Fig.A1 and Fig.A2).

12: **Else**

- 13: $\chi^{(k-1)} \leftarrow \Theta$

Implement P_A to select the most isolated point, c.f., Fig. A.3.

14: **End if**

OUTPUT

- 15: The most isolated point $x_c^{(k)}$.
-

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